

Gesture-Controlled Presentation System 🎮👐

Operate your Google Slides presentations using simple hand gestures captured through your laptop's camera. This project combines Computer Vision and AI technologies to create a touch-free presentation control experience.

The application leverages:

- OpenCV
- MediaPipe
- PyAutoGUI

to recognize hand gestures and translate them into presentation commands such as:

- Move to the Next Slide
- Return to the Previous Slide
- Launch Slideshow Mode
- Exit Slideshow Mode



Features

- ✓ Navigate Google Slides through hand gestures
- ✓ Real-time hand detection and tracking
- ✓ Uses your laptop's built-in webcam
- ✓ No external sensors or devices required
- ✓ Excellent beginner project for AI and Computer Vision enthusiasts



Gesture Controls

Gesture	Action
👐 Open Palm	Next Slide
🖐 Three Fingers	Previous Slide
✌ Two Fingers	Start Slideshow
✊ Fist	Exit Slideshow

Technologies Used

- Python
 - OpenCV
 - MediaPipe
 - PyAutoGUI
-

Installation

Step 0 - Verify Python Installation

For Linux and macOS users, replace `python` with `python3` in the commands below. </br>

Open the VS Code Terminal (Terminal → New Terminal) and verify Python is available:

```
python -V
```

Step 1 — Download the Project

```
https://github.com/shivam-kotwalia/gesture-control/archive/refs/heads/main.zip
```

Step 2 — Install Required Packages

```
python -m pip install -r requirements.txt
```

Step 3 — Download the Hand Landmarker Model

```
python setup_models.py
```

Launch the Application

```
python main.py
```

macOS Permission Setup

To allow keyboard automation on macOS, enable the necessary permissions.

Navigate to:

System Settings → Privacy & Security

Grant access to:

- Accessibility
- Input Monitoring

Allow permissions for:

- Terminal OR
- VS Code OR
- PyCharm

Without these permissions, keyboard commands cannot be sent to Google Slides.

Usage Instructions

1. Open your presentation in Google Slides using Chrome.
 2. Start slideshow mode.
 3. Launch the Python application.
 4. Position your hand in front of the webcam.
 5. Use the supported gestures to control the presentation.
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Project Structure

```
gesture-control/  
|  
├── .gitignore  
├── main.py  
├── README.md  
├── requirements.txt  
├── setup_models.py  
└── models/  
    └── hand_landmarker.task
```

requirements.txt

```
mediapipe==0.10.35  
opencv-python>=4.8.0  
pyautogui>=0.9.54
```

Working Principle

The system performs the following steps:

1. Captures live video from the webcam using OpenCV.
2. Detects hand landmarks with MediaPipe.
3. Analyzes finger positions and hand configurations.
4. Matches recognized gestures to predefined keyboard actions.
5. Sends keyboard inputs through PyAutoGUI to control Google Slides.

Future Enhancements

- Swipe-based slide navigation
- Gesture-controlled laser pointer
- Volume adjustment controls
- Zoom-in and zoom-out gestures
- Custom AI gesture training
- Multi-hand gesture support

Learning Outcomes

This project provides practical exposure to:

- Computer Vision
 - AI-powered Gesture Recognition
 - Human-Computer Interaction (HCI)
 - Real-Time Video Processing
 - Python-Based Automation
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Demo Applications

This project can be showcased during:

- AI Workshops
 - Hackathons
 - College Technical Festivals
 - Computer Vision Training Sessions
 - Smart Classroom Demonstrations
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Notes

- Use the application in a well-lit environment.
 - Keep your hand clearly visible to the camera.
 - Minimize background distractions for improved accuracy.
 - Best results are achieved when maintaining a moderate distance from the webcam.
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Live MediaPipe

MediaPipe Hand Tracking Demo

Google AI MediaPipe

Built with AI and Computer Vision

An intelligent gesture-based interaction system powered by real-time hand tracking, computer vision techniques, and AI-driven gesture recognition.